

EAGLE, AK, USAWMO: **701975**Lat: **64.777N**Lon: **141.148W**Elev: **908**StdP: **14.22**Time Zone: **-9.00 (NAK)**Period: **98-19**WBAN: **26422****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-49.5	-43.5	-35.4	0.8	-27.5	-33.5	0.9	-26.4	20.2	-10.5	18.5	-7.5	0.4	180	0.682

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.1	81.4	61.0	78.3	59.2	74.8	57.7	63.4	76.2	61.4	74.0	59.7	71.0	5.6	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.1	77.3	65.2	56.6	70.5	63.8	54.5	65.4	62.9	29.1	76.3	27.8	73.7	26.5	70.9	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	13.2	11.7	-50.8	86.5	5.9	2.6	-55.0	88.4	-58.5	89.9	-61.8	91.3	-66.1	93.2		
(5)			-33.3	66.7	2.8	4.5	-35.3	70.0	-36.9	72.6	-38.5	75.1	-40.6	78.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.7	-5.7	1.5	8.0	29.3	46.1	56.7	59.1	52.8	42.6	25.8	4.8	-1.5		
	DBStd	26.75	20.46	18.40	15.70	11.43	7.87	5.43	4.52	6.28	6.95	10.43	16.04	17.59		
	HDD50	9170	1727	1359	1303	622	172	8	3	42	232	750	1356	1596		
	HDD65	13972	2192	1779	1768	1072	587	253	188	380	671	1215	1806	2061		
	CDD50	685	0	0	0	0	50	209	285	129	12	0	0	0		
	CDD65	10	0	0	0	0	0	4	5	1	0	0	0	0		
	CDH74	844	0	0	0	0	83	301	323	137	0	0	0	0		
	CDH80	144	0	0	0	0	11	51	54	28	0	0	0	0		
	WSAvg	3.5	4.0	3.8	3.0	3.6	4.0	3.6	2.9	2.7	3.1	3.2	3.8	4.0		
	PrecAvg	11.7	0.5	0.4	0.3	0.4	0.9	1.8	2.2	1.9	1.2	1.0	0.7	0.7		
(15) Precipitation	PrecMax	15.7	1.7	0.9	1.5	0.8	1.9	3.4	4.8	3.9	3.3	1.9	2.2	2.0		
	PrecMin	6.7	0.1	0.0	0.0	0.0	0.0	0.4	0.6	0.6	0.0	0.1	0.0	0.0		
	PrecStd	2.6	0.4	0.2	0.3	0.2	0.6	0.8	1.0	0.9	0.7	0.4	0.5	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.2	44.1	48.3	65.3	81.6	84.8	84.7	83.8	69.1	53.0	41.5	41.6		
		MCWB	36.3	35.5	38.9	50.4	56.4	62.4	62.3	62.1	53.7	43.2	35.9	35.2		
	2%	DB	34.9	37.3	43.7	57.6	75.8	81.0	81.2	78.0	64.4	48.0	34.4	35.6		
		MCWB	30.8	31.2	35.1	43.4	54.7	60.8	61.2	61.5	51.2	40.0	30.7	31.0		
	5%	DB	27.0	33.1	38.3	53.4	71.7	77.4	77.8	72.7	61.5	44.9	30.4	28.3		
		MCWB	24.0	28.7	31.5	41.1	53.1	58.1	60.0	58.7	49.9	38.3	27.5	25.2		
	10%	DB	20.3	27.2	33.0	49.3	66.1	74.0	74.2	68.2	58.2	40.8	26.7	22.5		
		MCWB	18.2	23.7	28.0	38.8	50.4	56.3	58.9	56.8	48.5	35.9	24.2	20.3		
	0.4%	WB	37.6	36.1	40.3	50.0	60.4	70.8	65.6	67.6	55.0	44.5	36.7	35.8		
		MCDB	42.5	43.5	47.7	63.7	73.3	78.6	79.2	73.4	64.9	50.8	41.6	41.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	31.7	32.0	36.0	44.6	56.4	62.7	63.0	62.4	52.8	41.0	31.3	31.3		
		MCDB	36.4	37.3	43.1	57.1	71.8	76.6	75.7	75.4	61.7	46.7	34.1	35.6		
	5%	WB	25.9	28.9	32.1	41.5	54.0	59.9	61.6	59.9	51.0	38.4	27.8	25.9		
		MCDB	28.6	33.4	38.3	52.1	68.8	72.7	73.8	70.7	59.4	44.1	30.7	29.1		
	10%	WB	19.7	24.5	28.1	39.1	51.3	57.8	60.1	57.6	49.5	36.3	24.4	21.2		
		MCDB	21.7	27.9	33.1	48.4	65.2	70.7	71.5	66.3	57.7	40.9	26.8	23.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.7	21.0	29.0	28.4	29.5	29.1	26.1	25.8	23.6	16.1	14.6	15.2		
		MCDBR	19.0	23.8	30.1	32.7	39.8	36.8	35.5	35.4	31.3	20.1	15.2	17.0		
		MCWBR	16.4	18.7	22.4	20.0	20.5	18.9	18.0	19.9	19.5	13.8	12.8	14.2		
		MCDBR	19.5	23.0	27.8	31.1	36.4	31.2	29.5	30.8	27.6	18.4	15.0	17.0		
	5% WB	MCWBR	16.8	18.3	20.4	19.8	20.2	17.1	16.3	18.3	17.8	13.2	12.7	14.3		
(40) Clear-Sky Solar Irradiance	taub		0.189	0.245	0.254	0.326	0.354	0.384	0.400	0.370	0.318	0.263	0.171	0.174		
	taud		1.915	2.199	2.284	2.174	2.370	2.310	2.292	2.391	2.508	2.414	1.955	1.931		
	Ebn at Noon		108	214	269	266	271	263	253	251	245	210	117	39		
	Edh at Noon		8	17	25	36	33	36	35	29	21	14	7	3		
(44) All-Sky Solar Radiation	RadAvg		49	271	850	1512	1712	1767	1485	1175	733	310	76	15		
	RadStd		4	20	48	88	106	157	114	109	68	37	7	1		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	+4.95	+5.71	N/A	N/A	N/A	N/A	N/A	+99	N/A

Nomenclature: See separate page